



USER REQUIREMENT SPECIFICATIONS

ICOMS 2.0

Module 2 : Requestor and Planner

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1.0 DOCUMENT PURPOSE

The purpose of this document is to define the **User Requirements Specification (URS)** for the Requestor and Planner functionalities within the ICOMS 2.0 system.

This document describes the system behaviour, workflows, and functional requirements related to the **creation, management, and monitoring of outages** by Requestor and Planner users. It outlines the user interfaces, operational processes, and system capabilities required to support outage request submission, review, confirmation, change management, and reporting.

The document serves as a reference for system development, configuration, and validation to ensure that the ICOMS 2.0 system supports the operational needs of Requestor and Planner users in managing outage activities within the network.

2.0 OBJECTIVES

The objective of this document is to define the functional requirements that enable Requestor and Planner users to create, manage, and monitor outages within the ICOMS 2.0 system.

The key objectives include:

1. To define the system functionality required for creating and managing outages by Requestor and Planner users.
2. To establish the outage workflow and status progression from creation through agreement, confirmation, approval, and closure.
3. To describe the processes for reviewing, agreeing, and confirming outage requests prior to submission for GNM approval.
4. To define the functionality for editing outages and submitting change requests during the outage lifecycle.
5. To provide capabilities for viewing outage information, equipment history, and related operational data during outage planning.
6. To support monitoring, reporting, and repository access for outages within the system.

3.0 SCOPE

This document covers the Requestor and Planner user functionalities within the ICOMS 2.0 system.

The scope includes the following functional areas:

1. **Outage Creation and Submission**
Creation of planned, unplanned, and emergency outages including entry of outage details, equipment selection, and scheduling.
2. **Outage Workflow Management**
Management of outage status progression including Draft, Pending, Agreed, Confirmed, and submission for GNM approval.

3. **Outage Review and Confirmation**
Review of pending outages by Planner users and confirmation of agreed outages by Requestor users.
4. **Outage Monitoring and Repository Access**
Viewing outage listings, outage details, and system repositories containing outage information and status updates.
5. **Change Request Management**
Submission and tracking of change requests for outages pending approval.
6. **Information and Reference Access**
Access to supporting information including equipment history, maintenance history, project information, and conflicting line warnings.
7. **Reporting and Statistics**
Generation of reports and statistical views related to outages.
8. **Authorisation Monitoring**
Viewing outages that have been authorised and tracking their operational status through the system.

4.0 LIST OF USERS ROLES

No	Name	Division
1	Requestor	
2	Planner	
3		
4		
5		
6		

5.0 USER REQUIREMENT

5.1 Menu Update

Legacy	ICOMS 2.0	Remark
CONTROLLER MODULE		
Outage Pending Review	Outage Pending Review	Pending Agreement by Planner. Pending + Unconfirmed
Outage Pending Approval	Outage Pending Approval	Pending Approval by GNM. Agreed + Confirmed
REQUEST MODULE		
Planned Outage	Outage	Create New outage
Edit Planned Outage	Edit Outage	Edit created outages of the same Zone
Unplanned Outage	Removed	
Edit Unplanned Outage	Removed	
Emergency Outage	Removed	
Edit Emergency Outage	Removed	
OUTAGE STATUS		
Data Repository	Data Repository	
Confirmation Page	Confirmation Page	Pending Confirmation by Requestor. Pending/Agreed + Unconfirmed
Change Request	Change Request	Submit Change Requests for Outages not yet Approved + more than number of days set by GNM ADMIN; from start of outage
REPORTING MODULE		
Customised Report Page	Reports	
Statistics Page	Statistics	
Calendar Function	Calander	
AUTHORISATION MODULE		
Authorisation in Force	Authorisation in Force	Outages with current Taken Status
Long Authorisation	Long Authorisation	Outages with Taken Status that are more than 7 days
Authorisation Repo	Authorisation Repository	All Authorisations in current system
SPECIAL PAGE		
PMU Installation Management	Removed	
ACCOUNT INFO		

	User Account	Defined in URS Document 1
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This shows the Menu options in ICOMS 2.0 for the User Role Requestor and Planner.

5.2 Outage Workflow

Outages are created by multiple different roles and users.

Planned, Unplanned and Emergency outages can be created by Requestor, Planner and GNM team. Forced outages are only created by the GNC personnel.

5.2.1 Planned Outages

1. The Requestor/Planner will fill out the request form, based on the date/time selected the system will automatically classify the outage as a Planned outage. Status is set to **Draft** until submitted. Once submitted, status is Pending (Requestor side).
2. Once submitted, the Planner must agree to the outage by setting the status as **Agreed**. If the outage is created by Planner or GNM, this step is skipped and the status is automatically set to **Agreed**.
3. Planner will have to agree to the outage by selecting one of two options. "Agree" or "Agree – Push to DSO"
4. If pushed to DSO user, they will need to agree before the status is set to **Agreed**. DSO role is based across all regions, which will cause the user to see all outages pushed to them. If the outage pushed to DSO is not agreed in seven (7) days, the outage will be automatically agreed. This will be shown in the detailed view of the outage as Agreed by system.
5. There will be a new attachment field called Contingency Plan, where the users can upload a pdf document. This will be downloadable by the user with access to this outage by the right-hand side additional details panel.
6. If the outage is created by the planner, before the status is set to agreed, the system will prompt the user to send the outage to DSO.
7. If the Planner Disagrees to the outage, they shift the status back to **Draft**. The Planner can contact the Requestor to make comments outside the system in this scenario.
8. After the Planner has set the status to **Agreed**, the Requestor must now set the status to **Confirmed** at least 2 weeks before the outage is to start. If left unconfirmed, the status will be changed to **Outage Closed – Unconfirmed** automatically.
9. If the status has been **Confirmed** and **Agreed**, the GNM status shifts to **Pending**. GNM will conduct a study of the outage and complete the relevant fields including Under-Study notes, Highlights, Remarks, and Off-Points.
10. Requestor, Planner, and GNM users may edit the outage until the status changes to **Pending** under GNM. This is done when the outage has been **Agreed** to and **Confirmed** by the Planner and Requestor. After the **Pending** status on GNM side, edits are only allowed by the Change Request feature.
11. During the **Pending** stage, GNM will have the option to notify GCU users of the upcoming outage. This is done by GNM pressing the "Notify GCU" option on the right-hand information panel. Once selected, GNM user will need to select of the GCU users registered in the system who are assigned to the same zone as the outage. At the same time, a notification will be sent to the requester (GD, GM, GCU) to syndicate with the selected GCU users on the work program & contingency plans (Grid/DN level).
12. The GCU users will have a listing screen of "Outage Acknowledgement". The GCU user can Agree or Disagree to the outage, with a comment stating their reasoning. This does not impact the workflow of the outage as the decision is made by GNM regardless. This

Acknowledgement will be shown in the detailed view of the outage for GNM, in the right-hand information panel. The acknowledgement will be show on the listing screen as Informed (no reply from GCU), Agreed or Disagreed.

13. A notification/reminder will be given to GCU Users 2 days before the outage date.
14. GNM must now set the status to **Under-Study, KIV, Approved** or **Disapproved**. KIV shifts the outage to the KIV list with the remarks and comments by GNM that the Requestor can see. Approved status allows the GNC to begin work on the outage. Disapproved status ends the outage, the Requestor will need to resubmit a new outage.
15. If the outage remains untouched by the GNM until 24 hours after the start date/time, the outage will be sent to the KIV list.
16. Outages in the KIV list must have a change request submitted by the Requestor to edit, which will set the status to **Pending** (GNM) if the change request is approved by GNM. Change requests can edit the Date/Time, Voltage, Equipment along with adding Additional Equipment. If approved by GNM, the changes are made on the outage while a history is saved.
17. GNM can set the outages in KIV to a **Disapproved** status upon further study.
18. Outages remaining in the KIV list for more than 1 month will automatically be given the status **Outage Closed – KIV Timeout**.
19. Outages that are Approved will now be worked on by GNC. GNC will authorize and un-authorize the outages to get the status **Taken-Active** and **Taken-Completed**. This will happen over the duration of the outage.
20. Once the work has been completed, GNC will change the status of the outage to **Outage Closed**. This marks the end of the outage.
21. GNC users may extend the outage duration where operationally required.
22. Once the work is done, GNC will shift the status to **Outage Closed**.
23. If GNC does not act on an Approved outage, and its outage dates have passed, the system will wait 5 days, and then the status will be changed to **Outage Closed – Not Taken**. The requestor of this outage will not be able to create new outages until they give the reason for the Not taken status.
24. GNC can also cancel an outage with the status **Outage Closed- Cancelled by GNC**. If this case, the Requestor will be able to edit this outage, with the same outage ID to set dates which are outside the Planned parameters. – This outage will show two statuses on the statistic page, showing up on the **Outage Close – Cancelled by GNC** list and the **Outage Closed** status list.

5.2.2 Unplanned Outage

Unplanned outage will follow the same process as the Planned outage. Only exception will be the **Outage Closed – Not Taken** will not allow Requestors to submit a new edited outage. The Requestor will have to start from scratch on another Unplanned Outage.

5.2.3 Emergency Outage

Emergency outages when created will automatically have the status of **Agreed** and **Confirmed** upon creation. They will be parked in the GNM docket for study with the status **Pending**. The outage will follow the same process as the Planned Outage from that point on with the same exception on **Outage Closed – Not Taken** as the Unplanned Outage.

Please refer to Appendix A – Flowchart file for the complete flow chart diagram.

Requestor	Planner	DSO	GNM	GNC
-----------	---------	-----	-----	-----

Draft				
Pending				
	Agreed			
	Disagree			
		Agreed		
		Disagreed		
Confirmed			Pending	
			Under-Study	
			KIV	
			Approved	
			Disapproved	
				Taken-Active
				Taken - Completed
				Outage Completed
				Outage Closed - Not taken
				Cancelled by GNC
Outage Closed - Unconfirmed			Outage Closed - KIV Timeout	

Table 5.2.3a – Summary of Status by User Roles

5.3 Outage Pending Review

When an outage is created and submitted, it is given the Pending Status. This will be a listing of all newly created outages with the Pending Status, that the Planner will need to Agree to. The Planner will be able to select specific checkboxes, or use the Select All option to select all Check Boxes and Agree to all outages at once. Planner will also be able to mass Disagree to outages as well, which set their status back to draft for the Requestor.

Alternatively, the Planner will also be able to view the complete outage in detail and manually change the status from Draft to Agreed.

Check All		Uncheck All		Agree Checked Items		Disagree Checked Items			
Confirmed?	ID	Start / End Date		Station	Equipment Info		Description	Action	
☐	☐	155203	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 5:00PM Code : U		BTRZ (AM-KLUM)	Voltage Level:132 132BTRZPULU1 Additional Equipment: N/A	Defect Correction DEFECT RECTIFICATION 1, HOTSPOT P1 AT TOP PHASE 132KV BTRZ-SPKC TOFF MCRS T31/25. (38 °C)	[Review]	
☐	☐	155204	Wednesday, Apr 15 2026 12:00AM Wednesday, Apr 15 2026 12:00AM Code : U		NUNI (AM-KLUM)	Voltage Level:275 132ABBANUNI1 Additional Equipment: N/A	Defect Correction HOSPOT RECTIFICATION T9, BOTTOM PHASE P3 PIC SYAFIQ - 0166671990	[Review]	
☐	☐	155207	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 6:00PM Code : U		BTRZ (AM-KLUM)	Voltage Level:132 132BTRZ-SPKC/MCRS2 Additional Equipment: N/A	Defect Correction DEFECT CORRECTION HOTSPOT AT T31/25. P1 HOTSPOT (38.7 °C) PIC - SYAFIQ 0166671990	[Review]	
☐	☐	155210	Tuesday, Jul 21 2026 8:30AM Tuesday, Jul 21 2026 6:00PM Code : U		SRDE (AM-KLUM)	Voltage Level:275 275KVSRLGNG1 Additional Equipment: 132NUNIOIM2,	Defect Correction OUTAGE QUAD TOWER 275KV SRDE-LGNG L1 / 132KV NUNI-IOIM L2 HOTSPOT (P3) PIC SYAFIQ - 0166671990	[Review]	
☐	☐	155227	Tuesday, Apr 7 2026 9:00AM Tuesday, Apr 7 2026 10:00PM Code : U		KCPK (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T1 Additional Equipment: N/A	Others Surge Arrestor Replacement Works (Initiatives SUBD GMKL 2026). Jobsheet DOMS SEL-26-03105. PIC GMKL Hilmi (011-10320261)	[Review]	
☐	☐	155228	Tuesday, Apr 14 2026 9:00AM Tuesday, Apr 14 2026 10:00PM Code : U		KCPK (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T2 Additional Equipment: N/A	Others SA replacement Works (Initiatives SUBD GMKL 2026) Jobsheet DOMS SEL-26-03108 PIC GMKL Hilmi (011- 10320261)	[Review]	
☐	☐	155243	Monday, May 11 2026 9:00AM Tuesday, May 12 2026 10:00PM Code : U		CBJN (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T3	Others Retest HV bushing CBJN T3 (Initiatives SUBD GMKL) Jobsheet DOMS PTJ-26-00126 PIC GMKL Hilmi (011- 10320261)	[Review]	

Figure 5.3a Listing Screen showing outages to be Agreed or Disagreed

The DSO will have a similar screen, only showing the outages pushed to them by the Planner.

5.4 Confirmation Page

After the Outage has been Approved by the Planner, it must now be Confirmed by the Requestor before being sent to the GNM docket for review and approval. Requestors have until 2 weeks before the start of the outage to Confirm the outage. If left unconfirmed, the outage will be automatically given the status Outage Closed – Unconfirmed.

The Requestor will be able to select one or all the outages by using the checkbox system and Confirm All. The Reject All button will delete the outage from the system, as it has not passed to the GNM section yet.

Check All		Uncheck All		Agree Checked Items		Disagree Checked Items			
Confirmed?	ID	Start / End Date		Station	Equipment Info		Description	Action	
☐	☐	155203	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 5:00PM Code : U		BTRZ (AM-KLUM)	Voltage Level:132 132BTRZPULU1 Additional Equipment: N/A	Defect Correction DEFECT RECTIFICATION 1, HOTSPOT P1 AT TOP PHASE 132KV BTRZ-SPKC TOFF MCRS T31/25. (38 °C)	[Review]	
☐	☐	155204	Wednesday, Apr 15 2026 12:00AM Wednesday, Apr 15 2026 12:00AM Code : U		NUNI (AM-KLUM)	Voltage Level:275 132ABBANUNI1 Additional Equipment: N/A	Defect Correction HOSPOT RECTIFICATION T9, BOTTOM PHASE P3 PIC SYAFIQ - 0166671990	[Review]	
☐	☐	155207	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 6:00PM Code : U		BTRZ (AM-KLUM)	Voltage Level:132 132BTRZ-SPKC/MCRS2 Additional Equipment: N/A	Defect Correction DEFECT CORRECTION HOTSPOT AT T31/25. P1 HOTSPOT (38.7 °C) PIC - SYAFIQ 0166671990	[Review]	
☐	☐	155210	Tuesday, Jul 21 2026 8:30AM Tuesday, Jul 21 2026 6:00PM Code : U		SRDE (AM-KLUM)	Voltage Level:275 275KVSRLGNG1 Additional Equipment: 132NUNIOIM2,	Defect Correction OUTAGE QUAD TOWER 275KV SRDE-LGNG L1 / 132KV NUNI-IOIM L2 HOTSPOT (P3) PIC SYAFIQ - 0166671990	[Review]	
☐	☐	155227	Tuesday, Apr 7 2026 9:00AM Tuesday, Apr 7 2026 10:00PM Code : U		KCPK (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T1 Additional Equipment: N/A	Others Surge Arrestor Replacement Works (Initiatives SUBD GMKL 2026). Jobsheet DOMS SEL-26-03105. PIC GMKL Hilmi (011-10320261)	[Review]	
☐	☐	155228	Tuesday, Apr 14 2026 9:00AM Tuesday, Apr 14 2026 10:00PM Code : U		KCPK (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T2 Additional Equipment: N/A	Others SA replacement Works (Initiatives SUBD GMKL 2026) Jobsheet DOMS SEL-26-03108 PIC GMKL Hilmi (011- 10320261)	[Review]	
☐	☐	155243	Monday, May 11 2026 9:00AM Tuesday, May 12 2026 10:00PM Code : U		CBJN (AM-KLUM)	Voltage Level:132 132/33kV 90MVA T3	Others Retest HV bushing CBJN T3 (Initiatives SUBD GMKL) Jobsheet DOMS PTJ-26-00126 PIC GMKL Hilmi (011- 10320261)	[Review]	

Figure 5.4a Listing Screen showing Outages to be Confirmed

5.5 Outage Pending Approval

Outages that have been Agreed by the Planner and Confirmed by the Requestor will be waiting for Approval or Disapproval by the GNM users. The Requestor and Planner users can view the outages pending Approval in this listing screen, though they are not able to make any edits to the listing.

☐ Check All ☐ Uncheck All ☐ Agree Checked Items ☐ Disagree Checked Items							
Confirmed?	ID	Start / End Date	Station	Equipment Info	Description	Action	
☐	155203	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 5:00PM Code : U	BTRZ (AM-KLUM)	Voltage Level:132 132BTRZPULU1 Additional Equipment: N/A	Defect Correction DEFECT RECTIFICATION 1, HOTSPOT P1 AT TOP PHASE 132KV BTRZ-SPKC TOFF MCRS T31/25. (38 °C)	[Review]	
☐	155204	Wednesday, Apr 15 2026 12:00AM Wednesday, Apr 15 2026 12:00AM Code : U	NUNI (AM-KLUM)	Voltage Level:275 132ABBANUN1 Additional Equipment: N/A	Defect Correction HOSPOT RECTIFICATION T9, BOTTOM PHASE P3 PIC SYAFIQ - 0166671990	[Review]	
☐	155207	Monday, Apr 13 2026 8:30AM Monday, Apr 13 2026 6:00PM Code : U	BTRZ (AM-KLUM)	Voltage Level:132 132BTRZ-SPKC/MCRS2 Additional Equipment: N/A	Defect Correction DEFECT CORRECTION HOTSPOT AT T31/25. P1 HOTSPOT (38.7 °C) PIC - SYAFIQ 0166671990	[Review]	
☐	155210	Tuesday, Jul 21 2026 8:30AM Tuesday, Jul 21 2026 6:00PM Code : U	SRDE (AM-KLUM)	Voltage Level:275 275KVSRLGNG1 Additional Equipment: 132NUNIOIM2	Defect Correction OUTAGE QUAD TOWER 275KV SRDE-LGNG L1 / 132KV NUNI-IOIM L2 HOTSPOT (P3) PIC SYAFIQ - 0166671990	[Review]	
☐	155227	Tuesday, Apr 7 2026 9:00AM Tuesday, Apr 7 2026 10:00PM Code : U	KCPK (AM-KLUM)	Voltage Level:132 132/33KV 90MVA T1 Additional Equipment: N/A	Others Surge Arrestor Replacement Works (Initiatives SUBD GMKL 2026), Jobsheet DOMS SEL-26-03105, PIC GMKL Hilmi (011-10320261)	[Review]	
☐	155228	Tuesday, Apr 14 2026 9:00AM Tuesday, Apr 14 2026 10:00PM Code : U	KCPK (AM-KLUM)	Voltage Level:132 132/33KV 90MVA T2 Additional Equipment: N/A	Others SA replacement Works (Initiatives SUBD GMKL 2026) Jobsheet DOMS SEL-26-03108 PIC GMKL Hilmi (011- 10320261)	[Review]	
☐	155243	Monday, May 11 2026 9:00AM Tuesday, May 12 2026 10:00PM Code : U	CBJN (AM-KLUM)	Voltage Level:132 132/33KV 90MVA T3	Others Retest HV bushing CBJN T3 (Initiatives SUBD GMKL) Jobsheet DOMS PTJ-26-00126 PIC GMKL Hilmi (011-	[Review]	

Figure 5.5a Listing Screen showing outages to be Approved or Disapproved

The GCU will have a similar listing, showing only the outages pushed to them by GNM or acknowledgement.

5.6 Create new Outage

The primary function of the Requestor and Planner modules is the creation of outages within the system. Requestors and Planners will be able to create Unplanned, Planned and Emergency Outages.

Field	Field Type	Remarks
Zone	Dropdown	Zones Created by Admin
Station	Dropdown	Stations within the selected Zone
Voltage	Dropdown	Voltage across the system
Equipment Type	Dropdown	Equipment types for the Voltage Selected
Line Filter	Dropdown	Single, Tee-Off, Quad
Equipment	Dropdown	All Equipment of the Voltage and Type selected within the Station
Additional Equipment Type	Dropdown	Equipment types for the Voltage Selected. Also have the All option to display all Equipment within the station
Additional Equipment	Multi-Select	All Equipment for the additional type selected. Users can multi-select multiple equipment across types
Outage Type	Dropdown	Dead, Live
PTW	Checkbox	This will notify GNM of daily Authorisation and Un-authorisation of outage
Date Start and Time	Calander, Dropdown	Default time set to 0800, dropdown will show increments of 30 minutes
Date End and Time	Calander, Dropdown	Default time set to 1700, dropdown will show increments of 30 minutes

Code (Outage parameters)	Text Field	Auto filled with P, U, E based on Date/Time selected
Sequence	Dropdown	Sequence created by Admin
Restoration	Dropdown	Restoration times created by Admin
Job Type	Dropdown	Job Types created by Admin
Project	Dropdown	Will only be visible if any job type except Maintenance is selected. Will show a list of projects created by the Project Engineer.
Description	Text Box	No character limit, open text box for users to fill as needed.
Information Sidebar		
SLD	Hyperlink	PDF opens in new tab for the SLD of the station selected
Conflicting Lines	Hyperlink	Shows links to outages that are conflicting with the current selected details
History	Hyperlink	Will open a pop-up of the outage history of the equipment selected. Clicking the outage ID will open the detailed view in new tab
Maintenance History	Hyperlink	Outage ID and Date/time of the last Maintenance of the equipment selected
Project	Hyperlink	List of all outages in the project selected. This will show the outage IDs, job type, and description. The Outage ID, when clicked will open the outage detailed view in a new tab.
PIC	Pop-up Form	The PIC will open a form to allowing users to enter the Name, Email Address and Phone Number of multiple PIC.
Contingency Plan	Hyperlink	Downloadable PDF format CP uploaded by DSO

Logic For Outage Creation

- Outage Code will be auto filled once the date and time are selected. Options will be Planned, Unplanned and Emergency. The short codes will be P, U and E respectively.
- The Project dropdown will be visible if any Job type except Maintenance is selected. If Maintenance is selected, the last Maintenance Outage on this equipment will be shown in the Information Sidebar.
- Each outage will have an Outage Class- Maintenance or Project (all work types except Maintenance). This will be a new filter for all listing screens.
- All outage timing will be set to 24 hours clock, and the default start time will be set to 0800, with the end time at 1700. Users will be able to edit the time fields as required. Time dropdown will show increments of 30 minutes
- Each outage will have multiple status fields shown as Planner/GNM/GNC.
- DSO users' agreement will show under the Planner filed as Agreed.
- Requestor and Planner users may edit their outages until the Requestor Confirms (Confirmed status) the outage. Once Agreed by Planners and Confirmed by Requestors, the outage will get the status of Pending under the GNM heading; (Confirmed/Pending/-)
- Once the outage reaches Pending under GNM, all changes need to be made using the Change Request form. Change Requests are available until the Approved status by GNM, further Change Request option for the outage is locked. To make further changes the Requestor will need to call GNM and request a manual change.
- Requestor can Cancel the outage using the Change Request form.

- GNC user will only Acknowledge the outages, their response will not have any effect on the status of the outage.
- Tooltips will be available to guide users on how to complete each field.
- Only Planned and Unplanned outages need to be Agreed and Confirmed by the Planner and Requestor. All Emergency outages will go to Pending under GNM immediately.
- All PIC added in the PIC form will be emailed once the outage is created, and will be emailed again during the "Update and Notify" conformation by user.
- Description Field should have no word limit, allow the user to enter information which will be shown in the same format. I.e., Paragraphs breaks, empty lines etc.
- Once a Station and Voltage is selected, the SLD pdf file will be available for download from the right-hand information sidebar.
- Conflicting Lines – Taking the data from the Conflicting Lines set up the by the GNM Admin, if a Line is selected that has an outage created on the same dates and time as the current one, with a conflicting Line, the system will show a warning on the right information sidebar. Users will still be allowed to create this outage, but there will be an additional warning pop-up which will need conformation by the user before the outage is created.
- Redundant Outages – If an outage already exists for the same equipment, job type, and date, the system will identify the request as redundant and prevent the creation of the outage. Same equipment with different job types will be allowed and will not be marked as redundant, these can be created at the same time as another outage for the equipment to allow engineers the chance at completing two jobs in one outage window.
- The redundancy checker will also look though the list of KIV outages, and if there is one that matches, it will suggest the user submit a change request for the KIV outage instead of creating a new entry. – This KIV list will be based on Organisation, not user role.
- As a Requestor or Planner, when creating a new outage, the Planned outage will limit the Zone to the users designated Zone, as per their profile. However, if the outage type is Unplanned or Emergency, the user can select all available regions from the dropdown.
- If there is an outage with the OUTAGE CLOSED – NOT TAKEN status in the users designated Zone, the users will not be able to create new outages until they fill out the form selecting the reason for the NOT TAKEN status on the outage. If the reason is not in the dropdown list, users can enter their own reason. A justification is necessary for the Zone's users to have access to the outage creation methodology.
- If a Planned outage has been Cancelled by GNC, the requestor will have the ability to resubmit the outage on different dates, with the same Planned status. This is done by allowing the Requestor to edit the old outage and resubmit with new dates and time. This will reset the approval process and will need to go through the Agreed and Confirmed verification by Planners. The original date/time will be stored in the history of the outage.
- For Job Type – Routine Maintenance, the system should show the last Routine maintenance that was conducted on that equipment, along with any current scheduled outages for Routine maintenance. The user will the decide on how to proceed. This information will be shown on the right Information Sidebar.
- When creating an outage, upon selecting the equipment, the users should be able to view a history of the outages on the equipment. There will be a link for Outage History in the Information Sidebar. By clicking on it, it will show a listing of all the outages the equipment has gone though, with the outage job type shown, and a clickable outage ID. Clicking this Outage ID which will open the detailed view of the previous outage in a new tab.

- Equipment maintenance history will be shown for the last 10 years. The equipment outage history will be shown for the last 5 years.

Zone *

Select zone

Voltage *

Select voltage

Line Filter

Select line filter

Additional Equipment Type

Select additional type

Outage Type *

Select outage type

Date Start *

Pick a date

Date End *

Pick a date

Code (Outage Parameters)

Auto-filled

Job Type *

Select job type

Description *

Enter detailed description of the outage request...

Station *

Select station

Equipment Type *

Select equipment type

Equipment *

Select equipment

Additional Equipment (Multi-Select)

Select additional equipment type first

☐ PTW (Notify TOMS of daily Authorization/Un-authorization)

Time Start *

08:00

Time End *

17:00

Sequence *

Select sequence

Restoration *

Select restoration time

Project *

Select project

Save as Draft

Submit Outage Request

Information

Single Line Diagram (SLD)

Conflicting Lines

Outage History

Last Maintenance

Person In Charge (PIC)

Figure 5.6a Outage Creation Form

5.7 Data Repository

This will show a listing screen of all outages within the Zone of the user account, that have an Agreed and Confirmed status by Requestor and Planner. This listing will show the status updates along with any comments by GNM, and status by GNC. The user will have access to filters to refine the listing screen to their preference.

Month: December Year: 2025 Keyword: Status: All Apply Filter Export to Excel Refresh Data					
Confirmed?	Start / End Date	Region / Station	Equipment / Description	Region / CPU / M/D/C Status	
☒	153283 Monday, Dec 1 2025 12:58AM Friday, Dec 5 2025 11:30PM	JOH1 / DBAY	By : Redwood bin Samad 132336V 98MVA T2 Additional Equipment: Job Description: Overhaul final, testing feeder 330v and ROP Stability. Outage requested by DN. Po. Tuo Ibrahim : #18752421	Agreed / Approved / Taken-Completed Highlight(s): DN Job Study Remark: TL(T1-T2)-13MVA	View
☒	139619 Monday, Dec 1 2025 9:06AM Tuesday, Dec 2 2025 5:00PM	JOH1 / CBRU	By : Redwood bin Samad Busbar Coplex Additional Equipment: 1325V Main Switch, 1325V Reserve Busbar, Job Description: Routine Maintenance: Routine Maintenance by SUBD & PROG. PIC Code 19005408, NCZAM 0137158196, AFDJ 0140624642, N code 01987734885. 1 Dec 0000 -2 Dec 0900 BC-MMS. -2 Dec 0900-1800 BC-MS	Agreed / Approved / Taken-Completed Highlight(s): 1) DN to be informed and to prepare CP-2 Subject PLTO normative (153396) Study Remark: TL CBRU-2500V n-1 rem BS load line CBRU	View
☒	150173 Monday, Dec 1 2025 9:06AM Tuesday, Dec 2 2025 5:00PM	JOH1 / TBRU	By : SHAHRUL FITRI BIN ANNUARUSADAT 132MA-KOTIRU1 Additional Equipment: Job Description: Testing: TMBT-71-0203 RMR RTU & RELAY PMU TBRU: END OF WARRANTY	Agreed / R/Wtd / Pending Highlight(s): Study Remark:	View
☒	150166 Monday, Dec 1 2025 9:06AM Tuesday, Dec 2 2025 5:00PM	JOH1 / TBRU	By : SHAHRUL FITRI BIN ANNUARUSADAT 132336V 98MVA T4 Additional Equipment: Job Description: Testing: TMBT-71-0203 RMR RTU & RELAY PMU TBRU: END OF WARRANTY	Agreed / Approved / Taken-Completed Highlight(s): 1) DN to be informed and to prepare CP-2) DN to ensure rem to loading -98% Study Remark: T3-T4-95MVA	View
☒	153308 Wednesday, Dec 3 2025 8:36AM Monday, Dec 5 2025 5:00PM	JOH1 / KTRB	By : Wan Famy Adi Bin Wan Ahmad 273KTRPMAY1 Additional Equipment: Job Description: Switch: TRANS DISSECTION: PATAH SUBSTATION END OF WASTYTIME: SWAPERS END OF DISSECTION: RTU 9	Agreed / R/Wtd / Pending Highlight(s): Inflow long link outage Study Remark:	View

Figure 5.7a Data Repository Listing Screen

Enhancements to the Listing Screen

- Show the Restoration Time
- Show the PIC list
- Keyword will be able to search through all data
- Add filters to Outage Code (U,P,E,F); Station; Equipment Type; Job Type
- Allow users to sort by Start Date and End Date

5.8 Change Request

Once the GNM status has changed to Pending, no more edits are allowed to the outage submitted. All new changes will need to be made by the user of a Change Request Form. This will allow users to request specific changes from the GNM users. If approved by GNM, the changes will be shown on the Outage.

Users will only be allowed to request changes for the following fields.

- Date
- Time
- Voltage
- Equipment
- Add Additional Equipment

Every approved change request will store a history for the original data before the CR was approved. This data will be stored and viewable by the REQUESTOR and GNM through the statistics module, along with showing the number of Change Requests submitted.

Change request can only be submitted before the Approved status by GNM. The Approved status means the outage will begin the next day; to make any changes at this point, the Requestor will have to call GNM and request a manual change.

A change request needs to be Approved or Rejected by GNM before another change request can be submitted for the same outage. Approving a Change request changes the data of the Outage and the workflow will proceed as normal. Rejecting a change request will revert the outage back to the REQUESTOR with the comments from GNM. REQUESTOR can now submit a new change request.

5.9 Customised Reporting Page

This page will allow users to generate their own listing screen, by selecting any available filter options available within the system. The generated listings can then be exported out as PDF and Excel files as reports for the users.

Users will be able to save their selected filters as a Favourite. This will enable them to generate reports with one click. Users are allowed to save multiple filters as favourites, tied to the individual account.



Figure 5.9a Generate listing based on filters

5.10 Statistics

REQUESTOR and PLANNER user roles will both be shown the same set of statistics. These will be set within the system by default and the users will be able to generate these charts by selecting the appropriate fields.

By Year/Month	Type of Chart	Type of chart
Year, Month	Outage summary showing total number of approved outages by Departments	Bar Chart
Month	Outage status summary by Departments showing the total number of outages along with which of those were Taken Completed, Not Taken and Taken Active.	Table
Month	Total number of Planned, Unplanned and Emergency outages registered within the system	Pie Chart
Year	Total number of Routine Maintenance outage types, by outages completed and pending	Bar Chart

5.11 Calendar

Each user will have a Calendar showing a 5-week duration at any time. By default, the calendar will show the current month, but the users will be able shift forward and backwards across the months and years. There will be an additional feature to allow the custom selection of any 5 consecutive weeks to display (e.g. 2nd week of Feb to 3rd week of March). Users can also export the calendar out to a pdf document.

By default, the calendar will show all outages during the weeks selected, within the zone of the user. The user can fine tune the display using the filters provided to display only the desired outages. These filters will include, Job Type, Equipment Type, Outage Code, Zones, Voltage and Status.

The displayed outages, if in the Authorisation state by GNC, will follow the colour code for the ongoing outages.

White/grey for Outages with no Authorisation. Green for outages with Authorisation and on time. Yellow for outages with Authorisation and Extended by GNC. Red for outages with Authorisation and an expired time.

OUTAGE INFO FOR : 136064
Update
Close

Status:
KIV
☐ Denied in Docket

Highlight(s):

Study Remark:

[\[Check Study Remark History\]](#)

Reviewed By: NOR ALHUDA BT MOHAMMAD ISHAK

Start Date: 01-Feb-2026 12:00 AM

End Date: 01-Mar-2026 12:00 AM

Requested By: MUHAMMAD ZULHUSNI BIN MOHD ZAIRONI / AD-C

Region: KLUM

Station: PTAI

kV level: 275

Main Type: Line

Equipment: 275PTAISLKS2

Add Type: -Select Additional-

Additional:

Sequence: Not Applicable.

Confirmed?: ☐

Weightage: 1 [\[Reassign Weightage\]](#)

Restoration: Not Applicable

CT Ratio:

☐ Daily PTW

Figure 5.11b Detailed view on Click

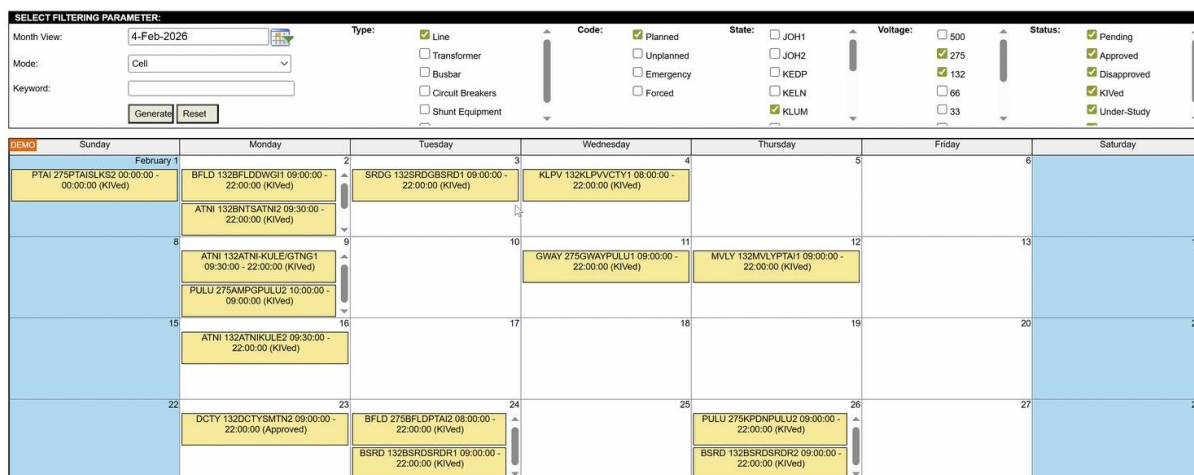


Figure 5.11c Calendar view by Cell

5.12 Authorisation in Force

Once Approved by GNM, outages can be worked on by the GNC team. This is done by Authorising the outage to give it a Taken-Active status. If the Authorisation is closed, the status changes to Taken-Complete. GNC can add and remove Authorisations on outages at will, as they work over the period of the outages.

The Authorisation in Force listing will show all the Outages that currently have a Taken-Active status.

The listing screen will be colour coded to show the timing of the outage. Green will be those are currently in the outage timeline window. Yellow will be for those that have been extended by GNC users. Red will denote the outages where the original time has been over, but they still have a Taken status, and have not been closed with any of the Outage Closed status.

Region	AB	Keyword	Month	Select Month	Year	Select Year	Sort	Start Date	ASC	Duration	Check	Cancel
Total No of Outage in Force found based on search criteria : 178 / Expired authorisation count as of 11/02/2026 : 148												
No	Order Info	Station / Stage	Region / Station	Equipment Info	Job Description	Action						
04079	Monday, Jan 5 2016 12:00PM Tuesday, Jan 12 2016 12:00PM Code: 1	980.5	KLUM / KLUM	275KV GCB 012 296 Additional Equipment 275KV GCB 012 296	Station : Taken-Active Disconnection of supply under SCADA TRIP (Operate W/ 21 instruction)	Review						
04186	Tuesday, Jan 12 2016 10:00PM Wednesday, Jan 21 2016 12:00PM Code: 1	1003.1	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						
04186	Tuesday, Jan 12 2016 10:00PM Wednesday, Jan 21 2016 12:00PM Code: 1	1003.1	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						
04147	Wednesday, Apr 23 2016 10:00AM Thursday, Dec 21 2016 12:00PM Code: 1	981.1	KLUM / SHDN	275KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired	Review						
71364	Thursday, Mar 9 2017 12:00AM Wednesday, Feb 28 2016 12:00AM Code: 1	156	KLUM / KLUM	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active Tripping	Review						
71364	Thursday, Mar 9 2017 12:00AM Wednesday, Feb 28 2016 12:00AM Code: 1	156	KLUM / KLUM	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active Tripping	Review						
71364	Thursday, Mar 9 2017 12:00AM Wednesday, Feb 28 2016 12:00AM Code: 1	156	KLUM / KLUM	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active Tripping	Review						
04185	Thursday, Aug 24 2017 3:00PM Monday, Dec 31 2016 9:30PM Code: 1	494.3	PHNG / GCB	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active Supply disconnection	Review						
04955	Sunday, Apr 15 2018 12:00AM Monday, Dec 31 2018 12:00AM Code: 1	200	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						
04955	Sunday, Apr 15 2018 12:00AM Monday, Dec 31 2018 12:00AM Code: 1	200	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						
04955	Sunday, Apr 15 2018 12:00AM Monday, Dec 31 2018 12:00AM Code: 1	200	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						
04955	Sunday, Apr 15 2018 12:00AM Monday, Dec 31 2018 12:00AM Code: 1	200	KLUM / SHDN	132KV GCB 012 296 Additional Equipment N/A	Station : Taken-Active SLA Expired, To disconnect generator	Review						

Figure 5.12a Listing screen of Authorisation in Force

Each Authorisation listing screen will have filters for Region, Month, Year, Start Date, On Time/Extended/Overdue and the ability to sort by Ascending or Descending based on start date.

5.13 Long Authorisation

Long Authorisation will show all Outages that have a duration of more than 7 days, that have at least one Taken-Active Authorisation. The list will show all both Taken-Active and Taken-Completed status. The listing screen will follow the same colour code and filter mechanism as the Authorisation in Force listing.

Users will be able to “View Authorisation Workplan” to show all the Authorisations and removal of Authorisations that have happened to date; along with the user who did the Authorisations, the time and the Authorisation Authority.

LONG AUTHORISATION (OUTAGE MORE THAN 7 DAYS)

Month:

-Select Month-

 Year:

-Select Year-

 Keyword: Org:

All

 Sort:

Date Start

ASC

Apply

WorkPlan

Excel

Total No of In-force Long Authorisation : 104

ID	Ath No	Date Info	Region / Station	kV	Equipment / Job Info	
[48879]	RISPB11811	Monday , Jun 9 2014 12:00PM Tuesday , Jan 31 2017 12:00AM Duration: 966 Days, 12 Hour(s)	TERG / TKLG (AM-TERG)	275	275PWSMTKL2 Additional Equipment: 275PWSMTKL2G1, Description: Disconnection of supply - under SGM TNBD (Operasi Wilayah 2) instruction.	[View Auth Workplan]
[81418]	RISPB15191	Tuesday , Jan 12 2016 10:00PM Monday , Dec 31 2018 12:00AM Duration: 1083 Days, 2 Hour(s)	KLUM / SRDN (AM-KLUM)	132	132kV GCB GT2 290 Additional Equipment: N/A Description: SLA Expired. To dismantle generator.	[View Auth Workplan]
[81418]	G302491	Tuesday , Jan 12 2016 10:00PM Monday , Dec 31 2018 12:00AM Duration: 1083 Days, 2 Hour(s)	KLUM / SRDN (AM-KLUM)	132	132kV GCB GT2 290 Additional Equipment: N/A Description: SLA Expired. To dismantle generator.	[View Auth Workplan]
[81417]	RISPB15470	Saturday , Apr 23 2016 10:00PM Monday , Dec 31 2018 12:00AM Duration: 981 Days, 2 Hour(s)	KLUM / SRDG (AM-KLUM)	275	275KV GCB GT3 M30 Additional Equipment: N/A Description: SLA Expired.	[View Auth Workplan]
[75364]	G266816	Thursday , Mar 9 2017 12:00AM Wednesday , Feb 28 2018 12:00AM Duration: 356 Days, 0 Hour(s)	KLUM / KULS (AM-KLUM)	132	132KULSNSPK1 Additional Equipment: N/A Description: Tripping	[View Auth Workplan]
[75364]	G322357	Thursday , Mar 9 2017 12:00AM Wednesday , Feb 28 2018 12:00AM Duration: 356 Days, 0 Hour(s)	KLUM / KULS (AM-KLUM)	132	132KULSNSPK1 Additional Equipment: N/A Description:	[View Auth Workplan]

Figure 5.13a Long Authorisation Listing Screen

5.14 Authorisation Repository

The Authorisation Repository will show a complete listing of all outages that have the status **Taken-Active** and **Taken-Completed**. The outages will leave this list when they are given any one of the **Outage Closed** status. Users will be able to “View Authorisation Workplan” to see the comments entered by the REQUESTOR with date and time tags. This workplan is also viewable by the GNM and GNC team. REQUESTORS are allowed to enter these details at any time during the outage until an **Outage Close** status is applied. The listing screen will follow the same colour code and filtering as the Authorisation in Force listing.

AUTHORIZATION LIST DATA												
Month: December		Year: 2025		Keyword:		Sort: Date Start		ASC		Apply Filter	PDF	Excel
ID	Date Info	Region/Station	Equipment Info	Job Info								
[151870]	Monday, Dec 1 2025 12:00AM Thursday, Dec 3 2025 6:00PM Duration: 2 Days, 17 Hour(s)	PPRG / SPID	132KV BSMVA T3 Additional Equipment: N/A	MLDC Status: Taken-Completed Work Description: SPID T3 Routine Maintenance (SRM 2022) Substation: Primary Routine Maintenance Protection Secondary Routine Maintenance POC: Naser 019-474 4480/ Almal 013-277 8997 View Auth WorkPlan								
[151870]	Monday, Dec 1 2025 12:00AM Thursday, Dec 3 2025 6:00PM Duration: 2 Days, 17 Hour(s)	PPRG / SPID	132KV BSMVA T3 Additional Equipment: N/A	MLDC Status: Taken-Completed Work Description: SPID T3 Routine Maintenance (SRM 2022) Substation: Primary Routine Maintenance Protection Secondary Routine Maintenance POC: Naser 019-474 4480/ Almal 013-277 8997 View Auth WorkPlan								
[151870]	Monday, Dec 1 2025 12:00AM Friday, Dec 5 2025 6:00PM Duration: 4 Days, 18 Hour(s)	PPRG / KLUM	132KV BSMVA T1 Additional Equipment: N/A	MLDC Status: Taken-Completed Work Description: To perform all replacement for KLUM T1. WPP test failed during routine maintenance. DEV & Monitor tests were monitored and decreased drastically over 3 months. WPP measured after removal of all bus T1, results satisfactory. Conclusion of testing and need to be replaced. POC: Dharun 0124567391 View Auth WorkPlan								
[152020]	Monday, Dec 1 2025 12:00AM Friday, Dec 5 2025 11:30PM Duration: 4 Days, 23 Hour(s)	JOH1 / DBAY	132KV BSMVA T2 Additional Equipment: N/A	MLDC Status: Taken-Completed Work Description: Testing feeder 35kV dan ROP Stability. Outage requested by DM. POC: Tuan Ibrahim, 0167752421 View Auth WorkPlan								
[152016]	Monday, Dec 1 2025 6:00AM Wednesday, Dec 3 2025 6:00PM Duration: 2 Days, 18 Hour(s)	NSEM / POPS	275PWSMTKL2 Additional Equipment: 132KV TRLAT1	MLDC Status: Taken-Completed Work Description: OUTAGE WORKS: Reconnect ACWS (117-736) - 67 MDS & (179-771) - 3 MDS. Contractor: BRUJAM SON DHO K17. Muhammad Dzul-Che Mustafa (019-394137) Competent Engineer: 1 En Mohamad (019-4318259) 2 En Dzulkef (019-711222) Broadband POC: M. K. Ponniah (012-862507) M. Ramana Kumar (012-3432921) View Auth WorkPlan								
[152016]	Monday, Dec 1 2025 6:00AM Wednesday, Dec 3 2025 6:00PM Duration: 2 Days, 18 Hour(s)	NSEM / POPS	275PWSMTKL2 Additional Equipment: 132KV TRLAT1	MLDC Status: Taken-Completed Work Description: OUTAGE WORKS: Reconnect ACWS (117-736) - 67 MDS & (179-771) - 3 MDS. Contractor: BRUJAM SON DHO K17. Muhammad Dzul-Che Mustafa (019-394137) Competent Engineer: 1 En Mohamad (019-4318259) 2 En Dzulkef (019-711222) Broadband POC: M. K. Ponniah (012-862507) M. Ramana Kumar (012-3432921) View Auth WorkPlan								
[152116]	Monday, Dec 1 2025 6:00AM Monday, Dec 8 2025 6:00PM Duration: 7 Days, 24 Hour(s)	JOH1 / GPTH	275PWSMTKL2 Additional Equipment:	MLDC Status: Taken-Completed Work Description: GPTH T2 Routine Maintenance (SRM 2022) Substation: Primary Routine Maintenance Protection Secondary Routine Maintenance POC: Naser 019-474 4480/ Almal 013-277 8997 View Auth WorkPlan								

Figure 5.14a Authorisation Repository

AUTHORISATION WORK PLAN				
Request ID: 48879				
Category of Defects: Non Critical				
Contractor: NA				
Defect Rectification: NA				
Remark: Both lines disconnected due to commercial issue Disconnection of supply - under SGM TNBD (Operasi Wilayah 2) instruction.				
Additional Info: 				
Status Update(s):				
No	Date Start		Date End	Status
1				
2				
3				
Previous Status Information:				
Date Start	Date End	Work Status	Updated By	Date of Update
09/06/2014	31/12/2016	both lines disconnected due to commercial issue	ISHAN BHARAT KUMAR PATEL	3/15/2016 2:20:00 PM
31/12/2016	31/12/2018	Based on TNBD-Asset Manager: Both lines disconnected due to commercial issue (under litigation at HQ level)	W.AMIRUL HAFIZ B. W. MD YUSOF	12/18/2017 12:33:00 PM
31/12/2016	31/12/2018	Based on TNBD-Asset Manager: Both lines disconnected due to commercial issue (under litigation at HQ level)	W.AMIRUL HAFIZ B. W. MD YUSOF	12/18/2017 12:34:00 PM
				Close

Figure 5.14b View Authorisation Workplan

6.0 USER SIGN OFF – Requestor - Planner

User #1 Signature: Name: Designation: Dept.: Date:	User #2 Signature: Name: Designation: Dept.: Date:
User #3 Signature: Name: Designation: Dept.: Date:	User #4 Signature: Name: Designation: Dept.: Date:
User #5 Signature: Name: Designation: Dept.: Date:	User #6 Signature: Name: Designation: Dept.: Date: